Remarks: (1) $T: V \to W$ and $S: W \to U$ are l. trans.

$\Rightarrow$ (then) $T \circ S: V \to U$ is a l. trans.

(2) $T: V \to W$

$T(\underline{0_V}) = \underline{0_W}$

(3) Suppose $B = \{\underline{v_1}, \cdots, \underline{v_n}\}$ is a basis of V and $T: V \to W$

$T(\underline{v_1}), \cdots, T(\underline{v_n})$

Specifically, (⊛! 仅对 linear trans. 成立)

$\forall v = \alpha_1 \underline{v_1} + \cdots + \alpha_n \underline{v_n} \in V$,

then $T(\underline{v}) = \alpha_1 T(\underline{v_1}) + \cdots + \alpha_n T(\underline{v_n})$

(证)

E.g., $\forall T: \mathbb{R}^n \to \mathbb{R}^m$ (A is a $m \times n$ matrix)

are of the form $T(\underline{v}) = \underline{A}(\underline{v})$

⊛ Consider a basis of $\mathbb{R}^n$

$\{\underline{e_1}, \cdots, \underline{e_n}\}$

By (3), $\forall T$, (Let $T(\underline{v_i}) = \underline{a_i}$) $\underline{a_1}, \cdots, \underline{a_n} \in \mathbb{R}^m$

(需求) $A := \left(\underline{a_1} \middle| \cdots \middle| \underline{a_n} \right)$

then $T(\underline{v}) = A\underline{v}$

E.g. 1: $T: \boxed{C^{\infty}(\mathbb{R})} \to C^{\infty}(\mathbb{R})$,

(证一 trans, 是否是 linear trans,)

$T(f)(x) := \int_0^x f(t)\,dt$

(a) $T(\alpha f + \beta g)$

$= \int_0^x (\alpha f + \beta g)(t)\,dt$

$= \alpha \int_0^x f(t)\,dt + \beta \int_0^x g(t)\,dt$

$= \alpha T(f)(x) + \beta T(f)(x)$

So T is l.

E.g. 3: $T: M_{n\times n}(\mathbb{R}) \to \mathbb{R}^1$,

$T(A) := A_{11} + \cdots + A_{nn}$

(Ex.: 证此 l.)

E.g. 4: $T: M_{n\times n}(\mathbb{R}) \to \mathbb{R}^1$

$T(A) : \det(A)$

T is not l. $(\Leftarrow \det(A+B) \neq \det(A) + \det(B))$

E.g. 5: $T: M_{m\times n}(\mathbb{R}) \to M_{k\times n}(\mathbb{R})$,

$T(A) := BA$ (B is a $k\times m$ matrix)

(Ex.: 证此 l. (用MAT2040))

Kernal / Image of $T: V \to W$

Def.: $\ker(T) = T^{-1}(\underline{0}) = \{\underline{v} \in V \mid T(\underline{v}) = \underline{0}\}$

$\operatorname{Im}(T) = T(V) = \{T(\underline{v}) \in W \mid v \in T\}$

(?) ← ~~(range(T))~~

E.g.1: $T: \mathbb{R}^n \to \mathbb{R}^m$, $T(\underline{x}) := A\underline{x}$

(解算ker/Im时当整体)

$\ker(T) = ---$

$= \{\underline{x} \in \mathbb{R}^n \mid A\underline{x} \overset{★}{=} \underline{0}\}$ ~~★~~ (!)

$= \underline{\operatorname{Nul}(A)}$ → (用MAT2040解方程)

$\operatorname{Im}(T) = \{\text{all possible outputs}\}$ ★

$= \{A\underline{x} \in \mathbb{R}^m \mid x \in \mathbb{R}^n\}$

$= \{x_1 \underline{a}_1 + \cdots + x_n a_n \mid \underline{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\}$

$= \underline{\operatorname{Col}(A)}$

E.g.2: $T: C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $T(f) = f'' + f$

(let $T_1(f) = f''$, $T_2(f) = \underset{(?)}{Id}(f) = f$

★ (both linear

$\Rightarrow T = T_1 + T_2$ is linear)

(zero func.)

$\ker(T) = \{f \in C^\infty(\mathbb{R}) \mid T(f) = f'' + f = \underline{0}\}$

(?) ← ~~(E.g~~ $\ker(T) = \{\alpha \sin(x) \overset{(?)}{+} \beta \cos(x) \mid \alpha, \beta \in \mathbb{R}\}$

(?) $= \operatorname{Span}\{\sin(x), \cos(x)\}$

Ex.: 证: If $T_1, T_2 : V \to W$ are l.,

then $T_1 + T_2$ is l.,

and αT_1 is l. $(\forall \alpha \in \mathbb{F})$

E.g. 3: $T : P(\mathbb{R}) \to P(\mathbb{R})$,

$T(p(x)) = p'(x)$

Prop.: $T : V \to W$ is linear,

then ① $\ker(T) \leq V$

and ② $Im(T) \leq W$

Proof ②:

Let $\underline{w_1} = T(v_1)$, $w_2 = T(v_2) \in Im(T)$

(WTS: $\alpha \underline{w_1} + \beta \underline{w_1} \in Im(T)$)

$\alpha \underline{w_1} + \beta \underline{w_2} = \alpha T(\underline{v_1}) + \beta T(\underline{v_2})$

$= \underline{T(\alpha \underline{v_1} + \beta \underline{v_2})} \in Im(T)$

Ramk / Nulity

Def.:

Theorem: $\operatorname{rank}(T) + n(T) = \dim(V)$ ($T: V \to W$ is L.)

?

E.g. 1: $T: C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $T(f) = f'' + 4f$

$\dim(\ker(T)) = \dim\{\underline{\alpha \sin(2x)} + \beta\cos(2x) \mid \alpha, \beta \in \mathbb{R}\}$

?

$n(T) = 2$

?

E.g. 2: $T: P(\mathbb{R}) \to P(\mathbb{R})$, $T(p) = p'$

$\dim(\ker(T)) = \dim\{\alpha \overset{1(x)=1}{\boxed{1}} \mid \alpha \in \mathbb{R}\}$

$= 1$

$\dim(\operatorname{Im}(T)) = \dim(\underline{P(\mathbb{R})}) = \infty$

? (not $\underline{\text{finitely generated}}$)

?

E.g. 3: $T: \underset{\dim = n+1}{P_n(\mathbb{R})} \to \underset{\dim = n}{P_{n-1}(\mathbb{R})}$, $T(p) = p'$

$\dim(\ker(T)) = 1$

$\dim(\operatorname{Im}(T)) = \dim\{P_{n-1}(\mathbb{R})\}$

$= n$

Prop 2.18: $T: V \to W$ is linear

① T is injective $\Leftrightarrow \ker(T) = \{\underline{0}_v\} \Leftrightarrow n(T) = 0$

$\hookrightarrow$ ※: If $T(\underline{a}) = T(\underline{b})$, then $\underline{a} = \underline{b}$

② T is subjective $\Leftrightarrow \mathrm{Im}(T) = W \Leftrightarrow \mathrm{rank}(T) = \dim(W)$

(② is always true)

Proof ①:

(証 ⇒)

Assume T is inj.

Suppose $\underline{v} \in \ker(T)$

(WTS: $\cancel{\forall}\,\underline{v} = \underline{0}_v$)

?

$T(\underline{v}) = 0_w = T(0_v) \Rightarrow \underline{v} = \underline{0}_v$

(定理①)

(所有 $\underline{v}$ (集合) 為 $\ker(T)$)

(証 ⇐)

Assume $\ker(T) = \{\underline{0}_v\}$

Suppose $T(\underline{a}) = T(\underline{b})$

$\Rightarrow T(\underline{a} - \underline{b}) = 0$

$\Rightarrow \underline{a} - \underline{b} \in \ker(T)$ $(= \{\underline{0}_v\})$

$\Rightarrow \underline{a} - \underline{b} = \underline{0}_v$

$\Rightarrow \underline{a} = \underline{b}$

$\Rightarrow T$ is inj.

isomorphism (同构)

Def.: Let V, W be v. spaces

$\underset{(?)}{\underline{V \cong}} W$ are isomorphic

$\Leftarrow$ if $\exists$ $\underset{(?)}{\underline{a}}$ $\underset{(?)}{\underline{\text{bijective}}}$ L. trans $T: V \to W$

(the $\underset{(?)}{\underline{\text{map } T}}$ is called; an isomorphism between V and W

E.g. 1: (见题)

E.g. 2: 找(有无): bijective L. trans.

$S: P_n(\mathbb{R}) \to P_{n-1}(\mathbb{R})$

(无) (?) (见题)

Prop. 2.18

③ $T: V \xrightarrow{\cong} W$ is a bij. L. trans.

$\Rightarrow$ We $\underset{(\exists)}{\text{have}}$ $\underline{\text{inverse } \underset{(?)}{\text{function}}}$. $T^{-1}: W \to V$

Proof:

Let $w_1 = T(\underline{v}_1)$, $w_2 = T(\underline{v}_2) \in W$

(WTS: $T^{-1}(\alpha \underset{[illegible]}{\underline{w_1}} + \beta \underline{w_2}) = \alpha T^{-1}(w_1) + \beta T^{-1}(w_2)$)

$T(\alpha T^{-1}(\underline{w_1}) + \beta T^{-1}(\underline{w_2}))$

$= T(\alpha \underline{v_1} + \beta \underline{v_2})$

$= \alpha T(\underline{v_1}) + \beta T(\underline{v_2}) = \underline{(?)}$

(习题)